

# Tianning Feng

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## SUMMARY

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UPenn CIS Master's student focused on AI agent applications, full-stack engineering, and machine learning systems. Approaches problems from user psychology and product thinking, turning cross-product capabilities into executable solutions. Skilled at building and refining agent workflows, and carrying products from zero to launch, validation, and iteration.

## EDUCATION

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**University of Pennsylvania**, MSE in Computer & Information Science 2025.08 – 2027.05

- Relevant Coursework: Artificial Intelligence, Database Systems, Embedded Systems.

**Emory University**, B.S. in Computer Science & B.S. in Psychology · GPA 3.794/4.0 2021.08 – 2025.05

- Honors: Dean's List 2024 Fall-2025 Spring, Dean's List 2024 Spring-2024 Fall.
- Relevant Coursework: Data Mining, Machine Learning, Data Structures & Algorithms, Cognitive Psychology.

## PROFESSIONAL EXPERIENCE

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**ByteDance · Data Platform** (Shanghai) – Agent Application Engineer Intern 2026.05 – 2026.09

- Started from user experience and needs, building, adapting, and validating dataset, ticket, and service-integration Skills to deliver end-to-end downstream platform capabilities; after these features and Skills shipped, downstream Skill call volume grew 520% in week two.
- Identified a user pain point and independently built the 0-to-1 pipeline from code changes and AI draft generation through human review to cross-repository display, then rebuilt repetitive records as weekly digest snapshots to improve information access.
- Drove the "Skill Management" initiative across teams from design to launch, supporting enable/disable, switching, and comparison across multi-source Skill versions; unified resolution and runtime validation ensure the agent only receives necessary, authorized Skills, trimming context and preventing disabled capabilities from being invoked by mistake.

**Liba Space** (San Jose, CA) – Full-stack Engineer Intern 2025.09 – 2025.12

- Owned and led development of the Resume Analysis & Rewrite feature from 0-1 for Jobnova, a North American job platform; seamlessly integrated it into the user flow, leading to a 50% increase in MAU.
- Collaborated with cross-functional teams to translate user needs into technical specifications and validation logic, leading the full development lifecycle from initial coding to production deployment.
- Optimized backend API workflows to reduce LLM token consumption and improve response quality; refined the system architecture based on user feedback and managed final quality assurance.

**Government-Sponsored AI Education Initiative** (Beijing, China) – Contract Product Leader 2025.05 – 2025.08

- Led the technical execution of a 0-1 RAG platform, managing the full development lifecycle from database schema design (SQL/Vector) to API integration for automated lesson planning.
- Enhanced system response quality by refactoring the retrieval layer with a hybrid BM25 + Vector approach, significantly boosting the relevance of retrieved academic contexts.
- Automated content generation workflows by building a parsing engine that transformed raw educational PDFs into structured formats, reducing manual PPT preparation time by 70%.

## RESEARCH EXPERIENCE

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**Melody Lab · Emory University** (Atlanta, GA) – Deep Learning Researcher 2023.09 – 2025.05

- Developed SeizureFormer, a novel Transformer-based time-series forecasting model in PyTorch to predict seizure risks using RNS data, achieving state-of-the-art performance in 1-14 day horizons.
- Engineered a dynamic hypergraph neural network for epidemic modeling, optimizing the message-passing algorithm to achieve 30% faster convergence and 7-12% higher accuracy than baselines.
- Authored 2 papers accepted at PSB 2026; SeizureFormer was selected for an oral presentation (Top 10%).

**Language Biomarker Lab · Emory University** (Atlanta, GA) – Research Assistant 2022.09 – 2024.11

- Reduced manual data processing workload by 80% by architecting automated ETL pipelines using Hugging Face and PyTorch for multi-modal data.
- Standardized 100+ GB of raw data into structured formats for downstream ML tasks, implementing efficient data loading strategies to optimize training throughput.

## PROJECT

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**TrainWeave** — Model Training Cloud Platform 2025.11 – Present

- Built a model-training specification middleware layer enabling seamless migration of training jobs between cloud training platforms and private clusters, with distributed compute scheduling.
- Selected for Wharton Venture Lab and awarded \$2,000 in seed funding; built in automatic parameterization and result-gating mechanisms to ensure model reproducibility and optimal cost-performance across heterogeneous compute.

**PhillyHome Insight** — Housing & Community Analytics Platform 2025.05 – 2026.01

- Gained 200+ early subscriptions by leading the full-stack engineering of an analytics platform that integrates heterogeneous datasets (property, census, crime) via SQL.
- Achieved sub-200ms query latency for complex geospatial queries by designing composite database indexes and optimizing RESTful API response structures.

**OpenAPS Simulation** — OpenAPS Simulation 2025.09 – 2025.12

- Achieved sub-100ms response latency in a safety-critical environment by optimizing C++ task scheduling and memory management on FreeRTOS.
- Engineered a reliable PID control loop to regulate insulin delivery based on real-time CGM sensor data, simulating medical-grade device standards.

**Smart Home Assistant Development** — Smart home assistant 2024.01 – 2025.01

- Enabled concurrent control of 15+ smart devices by architecting a scalable backend using MongoDB and Swift with real-time synchronization.
- Reduced data sync latency to under 200ms by implementing real-time database observers (Change Streams) and optimizing the data transmission layer.

## SKILL

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- **Engineering Skills:** Proficient in multiple programming languages (Python, TypeScript, SQL, etc.); skilled at taking products from 0 to 1 and iterating through operations; strong multilingual communication
- **Infrastructure & Tools:** Proficient with Docker, Git, AWS, MongoDB, Neo4j, Vector Databases, and similar tools; experienced in cross-platform / cloud development

## PUBLICATION

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- Feng, T., Ni, J., Gleichgerricht, E., Jin, W. SeizureFormer: A Transformer Model for IEA-Based Seizure Risk Forecasting. Pacific Symposium on Biocomputing (PSB 2026) · Oral presentation · Hawaii, 2026.
- Liu, S., Gong, S., Feng, T., Liu, Z., Lau, M. S. Y., Jin, W. Higher-order Interaction Matters: Dynamic Hypergraph Neural Networks for Epidemic Modeling (EpiDHGNN). Pacific Symposium on Biocomputing (PSB 2026).